

The Two Verbs — v1

A formalization of the noun/verb complementarity in the three-panel drawing

Companion to glide-from-the-stack v1–v2.1. All numerical claims verified (verify_two_verbs.py, 16/16 pass, tolerances as noted).

0. Provenance and aim

Source: the three-panel drawing (top view / front view). Panel 1: **depth lines** — noun: line objects, verb: covectors, field: affine slab, count scale 0–4. Panel 2: **true-length lines in the picture plane** — noun: line objects, verb: vectors, field: metric surface, ratio scale 0–1. Panel 3: **doubly-ruled surfaces** as combined pairs, captioned covector–vector, nouns–verbs / verbs–nouns.

Claim to be priced: *vectors and covectors can coexist in the same Euclidean space, but only in complementary form: noun with verb.*

Verdict, in advance: the claim is **theorem-grade after one repricing**. The coexistence is real and the complementarity is even stronger than the drawing asserts — under the apparatus group the two verbs are *mutually exclusive on a single line object*, not merely complementary. The repricing: the space in which this holds natively is not Euclidean 3-space but the apparatus space (affine slab $\times$ metric surface — the two field labels already on the drawing). Euclidean 3-space is one *completion* of the apparatus, and completing it dissolves the distinction. §3 and §4 make both halves precise.

1. Axioms (the apparatus)

A1 (Slab). An affine 3-space $\mathbb{A}$, with two distinguished parallel planes Λ_0 (front/picture plane) and Λ_1 (back plane).

A2 (Stack). The calibrated affine functional $h: \mathbb{A} \rightarrow \mathbb{R}$ with $h(\Lambda_0)=0$, $h(\Lambda_1)=1$. Its level planes are the stacked parallel planes of the drawing. (*As in glide-from-the-stack §1: the stack is the primitive; the grain §1.6 refines the calibration into the count scale 0–4 of the front view.*)

A3 (Level metric). A Euclidean metric g on the picture plane Λ_0 .

Lemma 1.1 (Propagation). g extends canonically to every level plane: translate Λ_0 to Λ_c by any affine translation; two such translations differ by an in-plane translation, which is a g -isometry, so the induced metric on Λ_c is independent of the choice. No connection, no Type A machinery, no perpendicular is needed — parallelism in the slab suffices. (*Theorem-grade; proof is the sentence above.*)

So the apparatus is exactly what the drawing's field labels say: a **metric surface** (each level plane, all carrying one metric) crossed with an **affine slab** (the calibrated depth direction, carrying only h). Nothing else is data. In particular: no metric across levels, no distinguished depth direction, no perpendicular.

2. The apparatus group

Theorem 2.1. The affine self-maps of $\mathbb{A}$ preserving the calibrated stack h and the level metric g are exactly, in adapted coordinates $(x, z) \in \mathbb{R}^2 \times \mathbb{R}$ with $h = z$:

$$(x, z) \mapsto (Rx + a + z \cdot b, z), \quad R \in O(2), \quad a, b \in \mathbb{R}^2.$$

Proof sketch. Preserving h pins the linear part's bottom row to $(0,0,1)$ and kills the z -component of the translation. Restricting to an isometry on each level forces the upper-left block into $O(2)$, constant in z by affineness. The remaining freedom is the in-plane translation a and the shear coefficient b — and the shear part is precisely the **glide flow**: G_λ of glide-from-the-stack §3 is the b -parameter subgroup. ■ (*Verified: checks A1–A2, error $\leq 9e-16$.*)

So the group is $O(2) \ltimes (\mathbb{R}^2_a \times \mathbb{R}^2_b)$: rotations/reflections of the picture, translations of the picture, and glides. **Glide projection is not merely compatible with the noun/verb structure — it is the structure's symmetry group.** The five-parameter identity component, extended by the Complement $z \mapsto 1-z$ (the affine engine's $\text{Stab}(\infty) = \{\text{id}, \text{Complement}\}$ of $v_2 \S(b)$) as an index-2 extension.

3. The Exclusivity Theorem (the heart)

What can the apparatus say about a single segment — what survives the group of Theorem 2.1?

Theorem 3.1 (Complete invariants). Let a segment have endpoints at levels $h(F)$, $h(P)$ with picture-plane shadow $s \in \mathbb{R}^2$ (top-view displacement).

1. The endpoint levels $h(F)$, $h(P)$ are always invariant. (*Check B3, error 0.*)

2. If $h(F) \neq h(P)$ (a non-level segment), the shadow is **pure gauge**: for any prescribed target shadow there is a glide shear achieving it, $b = (\text{target} - R \cdot s)/\Delta h$. The apparatus can say *nothing* metric about a non-level segment. Its entire invariant content is the pair of levels — i.e., the count. (*Check B1: arbitrary targets hit to 4e-15.*)
3. If $h(F) = h(P) = c$ (a level segment), the shadow's g-length is invariant, and (c, length) is a complete invariant. (*Check A2.*)

Corollary 3.2 (All depth lines are congruent). The apparatus group acts transitively on full depth lines $(F, P) \in \Lambda_0 \times \Lambda_1$. (*Check B2, error 8e-16.*) This upgrades the ledgered observation "every depth line, regardless of length or orientation, brackets proportional depths $0 \rightarrow 1$ " from observation to theorem: the group *identifies* them all. It is also why Panel 1 is drawn correctly in general position — wild angles, no common vanishing point. The wildness is gauge. There is exactly one depth line, seen through five parameters of gauge.

Corollary 3.3 (Exclusivity of the verbs). A single line object carries exactly one pure verb:

- **level $\Rightarrow$ vector.** Its count is trivially zero and its length is honest (the metric surface speaks).
- **non-level $\Rightarrow$ covector.** Its count is honest and its length is gauge (only the affine slab speaks).

There is no mixed case *for a single object*. The drawing's "only in complementary form: noun with verb" is therefore proved in a strengthened form: not just *paired*, but *exclusive* — the verbs partition the nouns. (Configurations of several objects do retain relative mixed invariants once one object's gauge is spent — that is where the pencil/ $\Delta\lambda$ /cross-ratio machinery already lives. The exclusivity is per-noun, not per-figure.)

The repricing. In genuine Euclidean 3-space the metric identifies vectors with covectors everywhere and Corollary 3.3 dissolves. The apparatus group of Theorem 2.1 is *not* a subgroup of the Euclidean group — glides are not isometries. The coexistence claim should read: vectors and covectors coexist as exclusive verbs **in the apparatus geometry**; Euclidean space is one gauge-completion of it (choose a depth metric, or equivalently a perpendicular $\Rightarrow$ the shears die $\Rightarrow$ the covector verb collapses into the vector verb). This is the Erlangen trade in one line: *the covector verb exists because the group is larger than Euclidean; the vector verb exists because it is smaller than affine*. The apparatus stands on **two rungs of the ladder at once — Euclidean in the picture plane, ratio/affine in depth** — which is precisely what the two field labels on the drawing declare.

4. The named geometry, and which side of it this is

4.1 The name. The structure (degenerate metric measuring in a plane-pencil of directions, one special linear measure transverse to it) is **simply isotropic 3-space**, one of the classical Cayley–Klein geometries — Strubecker’s isotrope Geometrie, the standard modern reference being Sachs, *Isotrope Geometrie des Raumes* (Vieweg 1990); the 2-dimensional version is Yaglom’s Galilean geometry. It is not exotic: Pottmann and Liu built planar-quad architectural meshes as principal meshes of surfaces in isotropic 3-space (2007), and Pottmann’s current work uses isotropic geometry for stress surfaces and structural design. The apparatus has literature — and the literature lives in architecture.

4.2 The transposition (this is the sharp part). Standard isotropic 3-space is presented *fiber-first*: a degenerate quadratic form $q(v) = |v_{xy}|^2$ on **vectors**, whose radical is a distinguished *direction* (the isotropic/fiber direction); its motion group G_6 is $(x, z) \mapsto (Rx + a, z + c_0 + c \cdot x)$ — rotations, translations, and *fiber shears* that tilt the level planes while fixing every vertical fiber.

The apparatus is presented *stack-first*: a distinguished **covector** dh (the stack) plus a metric on its kernel — equivalently a degenerate quadratic form on the *dual* space with radical $\langle dh \rangle$. Its motions (Thm 2.1) fix every level plane and tilt the fibers.

These are not the same subgroup of the affine group, and neither contains the other. They are **contragredient**: inverse-transpose maps the linear part of each exactly onto the linear part of the other. (*Check C, error 0. Also A3/A4: each group preserves its own degenerate structure and provably breaks the other’s.*)

Priced claim (theorem, verified at matrix level): the apparatus is isotropic geometry read on the covector side — dual isotropic space. The fiber presentation stabilizes a *point at infinity* (the isotropic direction); the stack presentation stabilizes a *plane-pencil with calibration*. Collection vs. evaluation, once again — and the transpose that carries one to the other is the same duality move as the engine’s glide $\leftrightarrow$ cross-slit transposition and v2.1 §8.3’s “stages dualize, plays don’t.” (*That last identification is priced as rhyme pending the 8.5’ check protocol; the contragredience itself is theorem.*)

4.3 Pending literature check. That Sachs’s G_6 is the fiber-preserving presentation is reconstructed here from the *i*-metric (*i*-distance = projected distance, invariant under $z \mapsto z + c \cdot x$) and confirmed numerically against that definition — but the convention should be checked against Sachs directly, per triangulation discipline.

5. The anchor (the compass exists here)

v2.1 §8.3’ found the missing piece of the anchorless duality story: a canonical anchor $\leftrightarrow$ a **marked polarity** $\leftrightarrow$ a marked conic — the compass. In isotropic geometry the compass is

native: the **i-sphere**, the paraboloid $x^2 + y^2 = 2z$, whose polarity Q implements the metric duality (i-distances of points $\leftrightarrow$ i-angles of planes) that Euclidean geometry lacks.

What the anchor actually does, computed (checks D1–D5, all exact):

1. **It normalizes the fiber group** — conjugation $g \mapsto Q g^{-T} Q$ maps G_6 to itself, swapping **picture translations** $\leftrightarrow$ **fiber shears** (with identical coefficients) and **inverting** the special (depth) translation. The compass's job is to rotate the group onto itself, exchanging the two kinds of "sliding" — exactly the §8.3' prediction that with an anchor in hand, duality exchanges collection $\leftrightarrow$ evaluation across the point/plane divide, never within one cast.
2. **The pole of the plane at infinity under Q is the fiber direction** (the paraboloid's axis) — "pole of $\ell_\infty =$ conic center," verbatim from §8.3', now with the center at infinity because the conic is parabolic.
3. **It does *not* carry the fiber presentation to the stack presentation.** Conjugating a glide shear by Q produces a projective map that moves the plane at infinity (bottom row acquires the shear coefficients) — it leaves the affine world entirely. The bridge between the two presentations is the *transpose* (§4.2), a stage-level $V \leftrightarrow V^*$ duality; the conic polarity is a play-level duality inside one presentation. Two different dualities, doing two different jobs. This cleanly repairs a conflation the engine has been circling since 8.5'.

Chiasmus, seated. The drawing's caption "nouns–verbs / verbs–nouns" now has two independent supports:

- *Incidence (observation-grade, elementary)*: in a doubly-ruled patch, each ruling family parameterizes the other — every line of one family meets every line of the other exactly once, so each family is the coordinate system (verb) for the other family's members (nouns).
- *Metric duality (theorem-grade at group level, §5.1)*: under the anchor, what acts as translation (vector verb) in one cast acts as shear/graduation (covector verb) in the dual cast, coefficients preserved. The chiasmus is the anchor's swap, drawn.

6. The pairing surface

The Type A patch $S(s,t) = (1-t) \cdot ((1-s)F_0 + sF_1) + t \cdot ((1-s)P_0 + sP_1)$ between two depth lines (ledgered: bilinear $\leftrightarrow$ doubly ruled; $T = 0 \leftrightarrow$ parallelogram $\leftrightarrow$ flat; sagitta $|T|/4$).

New content, in the two-verbs vocabulary:

1. **The two ruling families carry the two verbs** (Panel 3, proved): the level rulings lie in

stacked planes ($\Delta h = 0 \Rightarrow$ pure vector: honest, per-level lengths), the depth rulings are transversal ($\Delta h \neq 0 \Rightarrow$ pure covector: honest count, gauge shadow). A doubly-ruled patch is the *only* surface a single figure can rule with both verbs at once — it is the noun/verb pair made flesh. (Check E1: both families straight to 3e–16.)

- 2. **The patch is the graph of the pairing.** The level-ruling vector $L(t) = (1-t)L(0) + t \cdot L(1)$ is linear in the count t (check E2, 5e–16): the vector verb varies affinely along the covector graduation. Rank-one flavor — the surface reads as (transported vector) $\otimes$ (graduating covector) — priced as **rhyme** until a tensorial statement is fixed.
- 3. **Twist is invariant content.** $T = F_0 - F_1 + P_1 - P_0$ transforms by the rotation alone; the glide shear drops out (endpoint z 's are 0,0,1,1, so the shear contributes $(0-0+1-1)b = 0$). (Check E3.) So flat-vs-twisted is apparatus-invariant: the gauge that erases each depth line's individuality (Cor. 3.2) cannot erase the *relative* twist of a pair. First instance of §3.3's "configurations retain relative invariants."
- 4. **Conjecture (two slits = two verbs of one another):** the patch's two ruling families $\leftrightarrow$ the u/v channels; the two independent S_3 copies (on μ_u, μ_v) live one per family, and the seam $r_u = r_v$ is where the families' structures fuse. Falsifier: exhibit a channel quantity that does not restrict to its family's rulings. (Conjecture-grade; check protocol should ride along with 8.5'd's.)

7. Ledger

#	Claim	Price	Status
1.1	Level metric propagates canonically (translations are isometries)	theorem	proved
2.1	Apparatus group = $O(2) \ltimes$ (translations $\times$ glides); glide = the symmetry group of the two-verbs structure	theorem	proved + verified
3.1–3.3	Exclusivity: levels always invariant; non-level shadow is gauge; level length is honest; verbs partition the nouns	theorem	verified (B1–B3)
3.2	All depth lines congruent (upgrades the $0 \rightarrow 1$ bracket observation)	theorem	verified
—	"Coexist in the same Euclidean space"	repriced	coexist in the apparatus geometry;

			Euclidean completion dissolves it
4.2	Apparatus = dual (covector-side) isotropic geometry; groups contragredient	theorem	verified (C, A3–A4)
4.1/4.3	Identification with Sachs/Strubecker isotropic 3-space, fiber convention	observation	literature check pending
5.1	i-sphere polarity normalizes the fiber group, swapping translations↔shears, inverting depth	theorem	verified (D1–D3)
5.2	Pole of H_∞ = fiber direction (§8.3's anchor, made concrete)	theorem	verified (D5)
5.3	Polarity does NOT bridge the two presentations; the transpose does	theorem	verified (D4 + C)
6.1	Ruling families carry the two verbs, one each	theorem	verified (E1)
6.2	Patch = graph of the pairing (level vector linear in count)	theorem / rhyme (tensor reading)	verified / open
6.3	Twist defect is apparatus-invariant relative content	theorem	verified (E3)
6.4	Two ruling families ↔ two slits ↔ $S_3 \times S_3$	conjecture	protocol pending
—	Transpose-of-presentations = engine's glide↔cross-slit transpose	rhyme	rides on 8.5' checks

Struck at birth (so it doesn't enter the ledger wrong): "the i-sphere polarity converts glide to cross-slit." Computed false — the polarity conjugate of a glide shear is not affine (D4). The stage-level transpose does that work; the polarity works within a stage.

8. What the system now is, in one paragraph

A slab of affine space carrying one covector (the calibrated stack) and one metric (on the covector's own level planes). Its symmetry group is glide. Under that group every line object

speaks exactly one verb: level lines are measured, transversal lines are counted, and nothing is both. The geometry has a classical name — isotropic space — but read on the covector side; the transpose that flips it to the textbook presentation is the same duality that flips glide to cross-slit. The compass that Euclidean duality lacks exists here natively as the i-sphere, and its action is the chiasmus of the drawing's caption: it swaps which verb each motion speaks. And the doubly-ruled patch is the unique figure that holds both verbs in one surface — one ruling family measured, the other counted — with its twist defect as the first relative invariant the gauge cannot touch.

9. Next constructions

1. **Sachs convention check** (§4.3) — settle by literature, not reconstruction.
2. **Tensor statement for §6.2** — fix whether the patch is honestly $v \otimes dh$ in a coordinate-free sense, or only rhymes with it.
3. **Interactive: the-two-verbs-v1.html** — one figure, a gauge slider (R, a, b live), showing: a wild field of depth lines collapsing to congruent copies as the gauge acts (Cor. 3.2 made visible); a level line whose length readout never moves; a non-level line whose shadow readout spins freely while its count readout never moves; the patch with per-family verb readouts and the invariant twist number.
4. **Fold 6.4 into the 8.5' check protocol** — one falsifier list, run once.